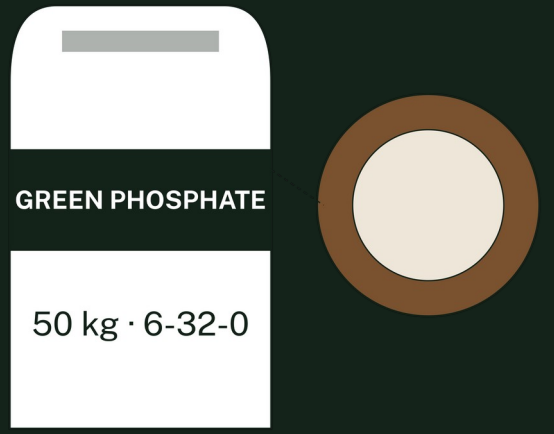
DEALER TRAINING · NOVEMBER 2026

Green Phosphate



Green Phosphate, 6-32-0 (6% nitrogen, 32% P_2O_5), is a phosphate granule made in an acidic mix and sealed with humic material. VAN makes it in Lahore in 35 kg and 50 kg bags.

VAN wrote this deck for its dealers and field team. A dealer can repeat its words to a grower.

THE 30-SECOND EXPLANATION

What a dealer says to a grower in 30 seconds.

Green Phosphate is 6-32-0, a phosphate granule made in an acidic mix and sealed with humic material, in 35 kg and 50 kg bags. On calcareous soil, which is above pH 8 and contains free lime, phosphate dissolving from a granule reacts with calcium within a few centimetres and slowly becomes unavailable to the root. Each Green Phosphate granule makes the soil around it acidic, which slows that reaction for the weeks a root needs to reach the granule. A 50 kg bag carries 16 kg of P_2O_5, against 23 kg in a 50 kg bag of DAP. Do not use the kilograms to call one bag better. It is a basal product: drill it at sowing, or broadcast it at land preparation where there is no drill.

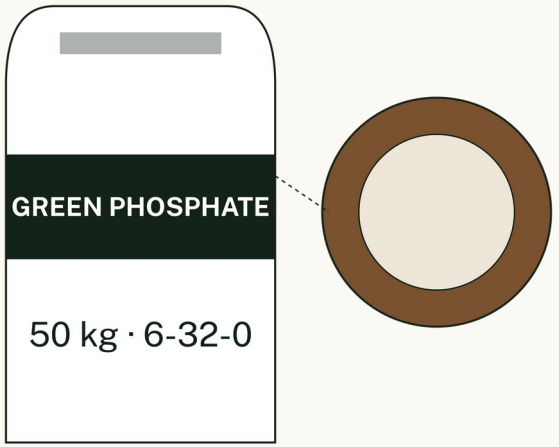
The 4 points it must cover

1. The bag: 6-32-0, 35 and 50 kg, PS 933-2026. A 50 kg bag carries 16 kg of P_2O_5 and 3 kg of N.
2. The soil: above pH 8 with free lime, phosphate reacts with calcium. In 35 of 36 Punjab districts, the median available phosphorus fell in VAN's Critical or Weak band (about 2016 to 2018).
3. How it works: each granule makes the soil within a few centimetres of it acidic for the weeks a root needs to reach it.
4. The stage: drilled at sowing, or broadcast at land preparation. Do not use the kilograms to call one bag better.

Explain the product in simple words a grower uses, and keep to the figures on these slides. Every figure in this 30-second explanation is on the bag, in the published crop plan or in the named evidence on the slides that follow.

THE BAG

What the bag declares, and how it is registered.



50 kg

bag weight. It carries 16 kg of P_2O_5 and 3 kg of nitrogen, by the declared analysis. The 35 kg bag carries 11.2 kg of P_2O_5 and 2.1 kg of nitrogen.

P_2O_5 32%

phosphate, declared on the front. In Urdu, the bag describes the product as a powerful acidic phosphatic fertiliser.

N 6%

nitrogen, declared on the front.

Acidic

granulated in an acidic mix, so the soil around each granule is acidic at first. Around a DAP granule, the soil is alkaline at first.

Humic

the seal on the granule: humic material that contains some fulvic material. No humic content figure is on the pack.

Basal

drilled at sowing, or broadcast at land preparation: 1 bag per acre on wheat and 2 bags per acre on potato.

Registered as	Pakistan Standard	Manufacturing licence
Green Phosphate	PS 933-2026	CM/L-2719/2023

If a grower asks about registration, give the licence number. Do not give an expiry date for the licence.

A GROWER'S QUESTION

“I apply a bag of DAP every year, and the soil test still says phosphorus is low. Where does the phosphate go?”

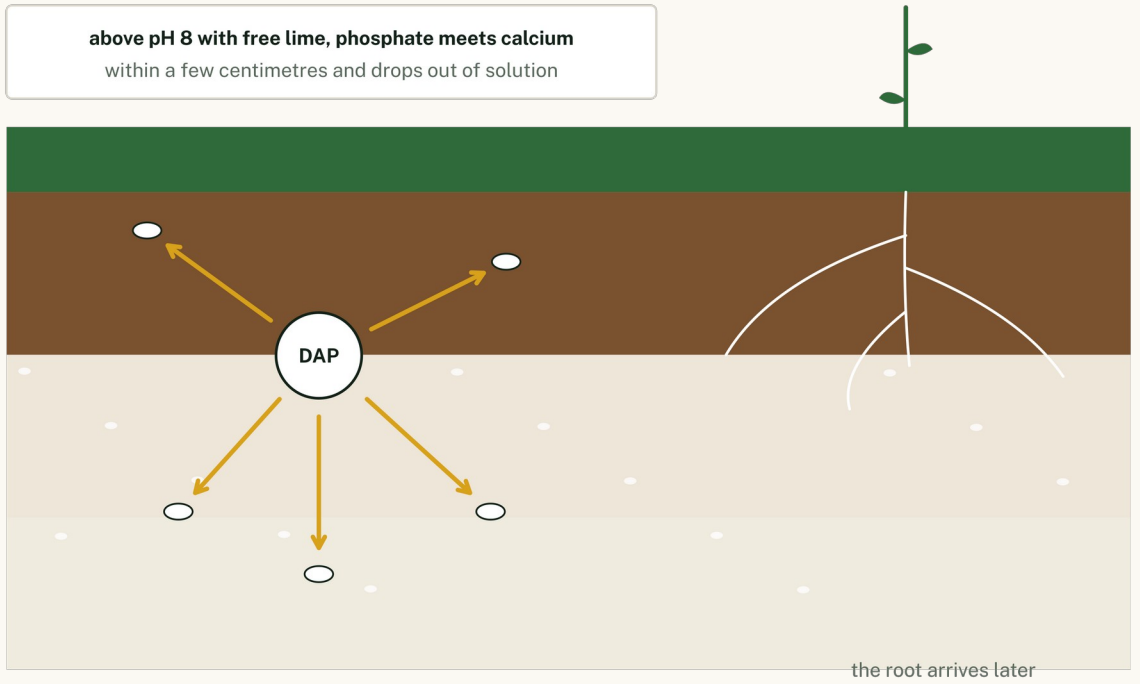
The Green Phosphate page of van.com.pk asks this question in the grower’s own words. The answer has 3 parts: what calcareous soil does to phosphate, what the acidic zone around each granule changes, and why the kilograms on the bag are not the right comparison.

The next 3 slides give the 3 answers, 1 idea on each slide.

ANSWER 1 OF 3

In calcareous soil, phosphate reacts with calcium before the root reaches it.

In soil above pH 8 that contains free lime, phosphate dissolving from a granule reacts with calcium within a few centimetres and comes out of solution as calcium phosphate. The first calcium phosphates that form are still fairly available to the crop. The phosphate becomes unavailable as these compounds slowly change into rock-like forms (apatite) that the root cannot use. In the Punjab soil survey (770,160 samples, tested in about 2016 to 2018), the median available phosphorus of 35 of the 36 districts fell in the Critical or Weak band of VAN's soil-test scale.



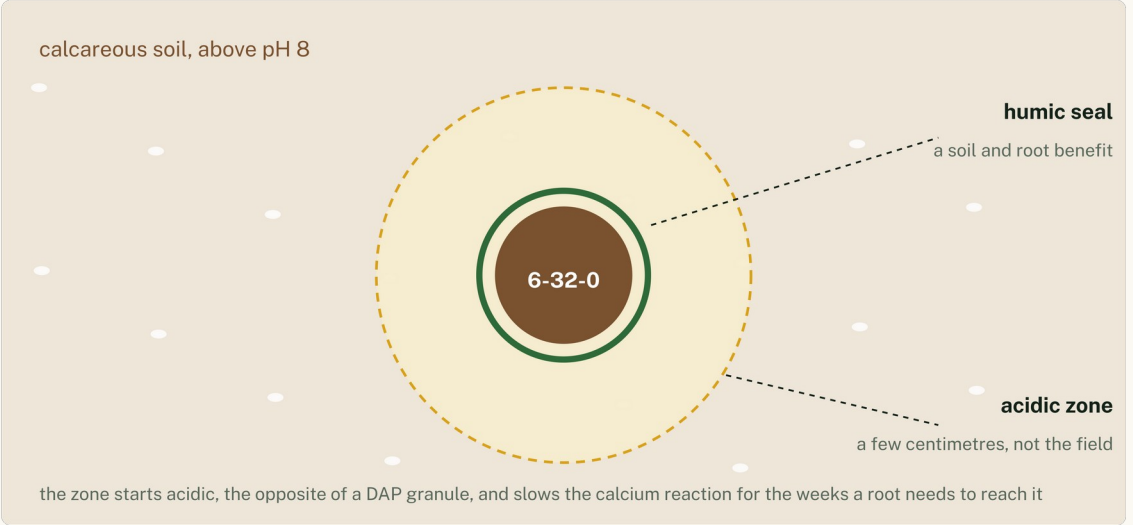
The stones in the drawing are free lime (calcium carbonate). The calcium reaches the phosphate before the root does.

Say it like this: "The phosphate reacts with calcium in the soil before the root reaches it. That is where much of the phosphate from your DAP went: it slowly became unavailable to the root."

ANSWER 2 OF 3

Each granule makes the soil around it acidic for the first few weeks.

Green Phosphate is granulated in an acidic mix and sealed with humic material that contains some fulvic material. The soil around each granule is acidic at first; around a DAP granule it is alkaline at first. The acidic zone slows the reaction with calcium during the weeks a root needs to reach the granule. The effect is limited to a zone a few centimetres wide, and it does not change a soil test result.



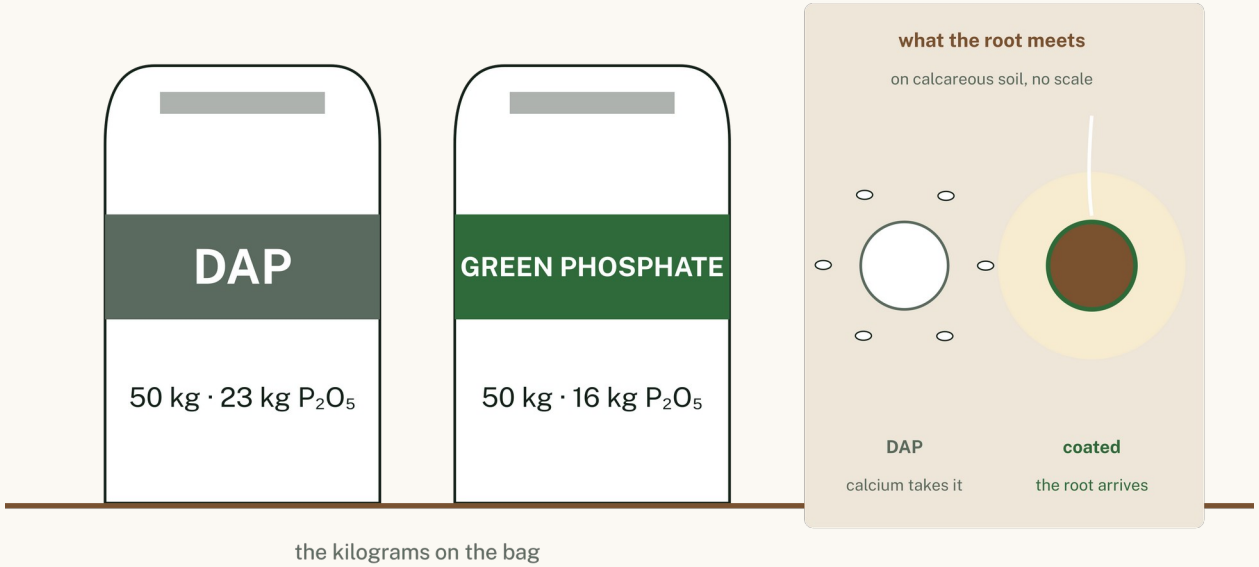
The acidic zone around the granule is what slows the reaction with calcium. The humic material is drawn as the seal; it does not shield the phosphate from calcium.

Say it like this: “Each granule makes a small acidic zone around itself, a few centimetres wide, for a few weeks. It does not change the whole field.”

ANSWER 3 OF 3

Do not use the kilograms to call one bag better.

A 50 kg bag of DAP carries more phosphate than a 50 kg bag of Green Phosphate. The crop can use only the phosphate that stays available in the soil. On calcareous soil this is much less than the amount in the bag, although VAN has no figure for it. A review of 81 published studies (Rose et al. 2014) found that humic substances increased shoot and root growth.



The drawing gives no availability percentage. The right-hand panel shows the idea and has no scale.

Say it like this: “The DAP bag carries 23 kg of P_2O_5 and the Green Phosphate bag 16 kg. What matters is how much of each the root can reach.”

THE EVIDENCE

What is verified for Green Phosphate, and what is not.

A dealer may state each item below only in the way the right-hand column allows.

Item	What it says	How a dealer may use it
The bag	GREEN PHOSPHATE, “multi nutrient fertilizer”, P_2O_5 32%, N 6%, 50 kg; in Urdu, a powerful acidic phosphatic fertiliser. Humic material is not named on the front of the bag.	As fact.
The licence	Pakistan Standard PS 933-2026, manufacturing licence CM/L-2719/2023.	As fact.
The soil chemistry (published)	Phosphate precipitates with calcium. The first calcium phosphates are still fairly available; the phosphate becomes unavailable as they slowly change into apatite, a rock-like calcium phosphate (Tunesi 1999; Freeman and Rowell 1981). In the Punjab soil survey (770,160 samples, tested in about 2016 to 2018), the median available phosphorus of 35 of the 36 districts fell in the Critical or Weak band of VAN’s soil-test scale.	As fact about the soil, never as a percentage of phosphate fixed.
Humic (published)	A review of 81 published studies (Rose et al. 2014): plants given humic substances had about 22% more shoot growth and about 21% more root growth than plants without them.	Spoken, as humic research. Never as a Green Phosphate result.
Not on file	A humic content figure; a percentage of phosphate saved or fixed; a soil-test change.	Not said.

Sources: VAN’s flagship audit, section 3c (items printed on the pack or verified); the description of the mechanism VAN approved on 27 September 2026; the regulatory table and the Green Phosphate page on van.com.pk. Pot studies of humic-coated phosphates and the withdrawn “70 to 80% fixed” line are not used.

Green Phosphate and DAP per bag: P_2O_5, nitrogen and price.

Prices are delivered farmer prices from the VAN farmer price sheet, August 2026. P_2O_5 and nitrogen are calculated from the declared analysis on each bag. No availability percentage is given, because none has been measured.

Bag	Weight	P_2O_5	Nitrogen	Price (Rs)
Green Phosphate, 6-32-0	50 kg	16 kg	3 kg	16,800
Green Phosphate, 6-32-0	35 kg	11.2 kg	2.1 kg	12,000
DAP, 18-46-0	50 kg	23 kg	9 kg	your counter price; not on the VAN price sheet

The DAP bag carries 23 kg of P_2O_5 and the Green Phosphate bag 16 kg. The comparison that matters is how much phosphate stays available to the root on calcareous soil. Slides 5 and 6 explain why. VAN has no figure for the share that stays available.

Green Phosphate 50 kg Rs 16,800 and 35 kg Rs 12,000: VAN farmer price sheet, August 2026. DAP analysis 18-46-0 from VAN's product master list; DAP has no price on the VAN farmer price sheet, so the grower writes in his own. 16 kg = 50 × 0.32; 23 kg = 50 × 0.46.

Green Phosphate per acre at land preparation, by crop.

Bags per acre are from the published crop plans, P_2O_5 from the declared analysis and prices from the VAN farmer price sheet, August 2026.

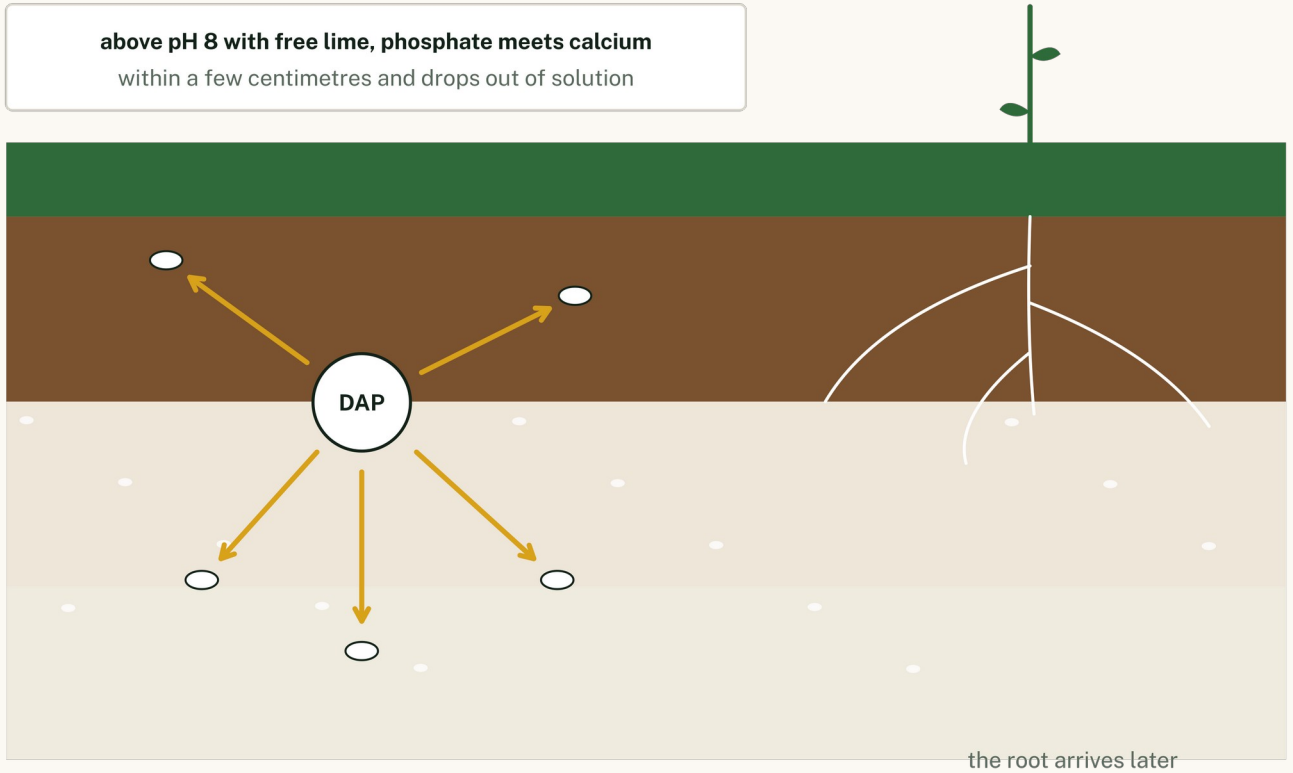
Crop	Green Phosphate per acre	P_2O_5 applied	Nitrogen applied	Rs per acre
Wheat	1 bag of 50 kg	16 kg	3 kg	16,800
Maize	1.5 bags of 50 kg	24 kg	4.5 kg	25,200
Cotton	1.5 bags of 50 kg	24 kg	4.5 kg	25,200
Potato	2 bags of 50 kg	32 kg	6 kg	33,600
Sugarcane	2 bags of 50 kg	32 kg	6 kg	33,600
Basmati rice	1 bag of 50 kg	16 kg	3 kg	16,800

For these crops, Green Phosphate is the only phosphate the crop plan applies at land preparation. The crop plan gives the same number of bags whether the grower uses Green Phosphate or Fusion Phosphate, so he can change the product without changing the plan.

The Green Phosphate rows of the published crop plans; 50 kg bag at Rs 16,800. Bag count on a switch of source: the Green Phosphate page on van.com.pk.

HOW IT IS APPLIED

A basal product: drill it at sowing, or broadcast it at land preparation.



Land preparation

Most crop plan rows apply Green Phosphate at land preparation, or at transplanting for basmati rice, tomato, chilli and onion. The tomato and banana plans also apply it at germination.

Drill or broadcast

Put whole granules into the soil, where the root will grow. Most rows say drill at sowing, or broadcast if there is no drill. Follow the method column of the crop plan.

A basal product

It does not correct a standing crop, and it is not a foliar product. The tomato plan gives fertigation as an alternative method, and the banana plans apply it by fertigation at some stages; use the method and rate in the plan.

OBJECTION 1 OF 8

“Rs 16,800 for a 50 kg bag, and it has less phosphate than DAP?”

What the dealer says

Yes, 16 kg of P_2O_5 against 23 kg. Do not use the kilograms to call one bag better. On soil above pH 8 with free lime, the phosphate from a DAP granule reacts with calcium within a few centimetres and slowly becomes unavailable to the root. The soil around a Green Phosphate granule is acidic at first, which slows that reaction for the weeks the root needs to reach the granule. Write both bag prices down, and then ask how much phosphate from each bag the root can reach.

The facts behind the answer

- Green Phosphate 50 kg Rs 16,800, 35 kg Rs 12,000 (VAN farmer price sheet, August 2026). DAP: the counter price, not on the VAN sheet.
- 16 kg P_2O_5 and 3 kg N against 23 kg and 9 kg, from the declared analyses.

Never give a percentage of phosphate saved or available. VAN has measured none.

OBJECTION 2 OF 8

“Fewer kilograms means less feed.”

What the dealer says

The bag carries less phosphate. It is made to slow the reaction with calcium near the granule, but VAN has no figure for how much more stays available. The crop can use only the phosphate that stays available in the soil. The crop plan does not add bags to make up the kilograms: it gives 1 bag of Green Phosphate on wheat and 2 on potato.

The facts behind the answer

- Green Phosphate page, answer 3: the crop uses only the phosphate that stays available.

Never say: “It beats DAP.” The 1 wheat trial covers 1 season on 1 farm, and TSP was not in it.

OBJECTION 3 OF 8

“I used it and my soil test did not change.”

What the dealer says

A soil test is not expected to change, and the Green Phosphate page says so. The acidic zone covers a few centimetres around each granule for a few weeks, while a soil test measures the whole field. Green Phosphate is designed to affect how much phosphate the root can reach in those weeks. It does not affect the soil test reading. The soil test stays low because the soil is calcareous, and Green Phosphate is made to form an acidic zone for that reason.

The facts behind the answer

- The Green Phosphate page lists a field-scale pH change, a soil-test change and a percentage of phosphate saved among the claims VAN does not make.
- In the Punjab soil survey (about 2016 to 2018), the median available phosphorus of 35 of the 36 districts fell in the Critical or Weak band of VAN’s soil-test scale.

Never say: “It raises your soil phosphorus” or “it changes the soil pH”.

OBJECTION 4 OF 8

“Can I put it through the drip?”

What the dealer says

Not for most crops. Green Phosphate is a basal granule: for wheat, maize, cotton, potato, sugarcane and rice, the crop plan puts it into the soil, drilled at sowing or broadcast at land preparation. Only the tomato and banana plans allow fertigation, at the rate in the plan. For every other crop, put it in the soil before the crop is sown or transplanted.

The facts behind the answer

- Methods in the crop plans: drill at sowing or broadcast if there is no drill; “Broadcasting/ Fertigation” on tomato; fertigation on banana.
- VAN’s flagship audit: basal, drilled at sowing or broadcast at land preparation.

Never give a foliar rate, and never recommend foliar use.

OBJECTION 5 OF 8

“At which stage do I put it on?”

What the dealer says

Put it in at land preparation or at sowing for every field crop, and at transplanting for basmati rice, tomato, chilli and onion. The young plant needs phosphorus for its roots in the first weeks, and the acidic zone around the granule lasts long enough for the young root to reach the granule.

The facts behind the answer

- The Green Phosphate rows of the published crop plans, 12 crop plans, on the crop plan slides.

Never say: “Top-dress it later if you forgot.”

OBJECTION 6 OF 8

“How many bags per acre?”

What the dealer says

Use the crop plan rate: 1 bag on wheat, rice, onion and citrus; 1.5 bags on maize and cotton; 2 bags on potato, sugarcane and chilli; on tomato, 1 bag at land preparation or transplanting and 1 at germination. The plan gives the same number of bags for Fusion Phosphate, so the grower can change the product and keep the plan, except on tomato: on rows that use fertigation, the plan offers no Fusion Phosphate. If he changes to Fusion Phosphate, he makes up 3 kg of nitrogen for each bag.

The facts behind the answer

- Rates per acre as published in the crop plans (crop plan slides).
- Bag count unchanged on a switch of source, except on rows that use fertigation: the Green Phosphate page on van.com.pk.

Never say: “½ bag will do because it is more available.”

OBJECTION 7 OF 8

“Is it registered? What is the licence?”

What the dealer says

It is registered under Pakistan Standard PS 933-2026, with manufacturing licence CM/L-2719/2023, and VAN is the registered maker. Every batch is tested at VAN Lab, accredited by the Pakistan National Accreditation Council (PNAC) as LAB 336, before it ships. On van.com.pk, enter the batch number from your bag, and you can read the certificate for that batch.

The facts behind the answer

- Regulatory table on van.com.pk: Green Phosphate, PS 933-2026, CM/L-2719/2023.

Never give a licence expiry date, or a humic percentage as if it were on the certificate.

OBJECTION 8 OF 8

“My crop is already standing and looks short of phosphorus. Will this fix it?”

What the dealer says

Green Phosphate is a basal product: it goes into the soil before or at sowing, where the root will grow. It does not correct a standing crop, and VAN does not sell it for that. Plan it for the next sowing. For the crop standing now, apply the product the crop plan gives for its next stage.

The facts behind the answer

- VAN’s flagship audit: basal, drilled at sowing or broadcast at land preparation.

Never say: “It fixes a standing crop” or “you will see it in a week”.

Green Phosphate in the published crop plans, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Land Preparation	1 bag	Drill at sowing; broadcast if no drill	50 kg
Maize (Corn)	Land Preparation	1.5 bags	Drill at sowing; broadcast if no drill	50 kg
Basmati Rice	Land Preparation/Transplantation	1 bag	Broadcasting	50 kg
Hybrid Rice	Land Preparation	1 bag	Drill at sowing; broadcast if no drill	50 kg
Cotton	Land Preparation	1.5 bags	Drill at sowing; broadcast if no drill	50 kg
Sugarcane, February planting	Land Preparation	2 bags	Drill at sowing; broadcast if no drill	50 kg
Ratoon Sugarcane	Land Preparation	2 bags	Drill at sowing; broadcast if no drill	50 kg

The rows are printed as published in the crop plans on van.com.pk. Coverage: these 2 slides show 12 of the 26 crop plans that name Green Phosphate, and 13 of its 33 rows. Of these 13 rows, 7 say drilling at sowing (or broadcasting), 4 say broadcasting and 2 say broadcasting or fertigation. Every row uses the 50 kg bag. Each rate is per acre, and the crop page scales it to the size of the field.

Green Phosphate in the published crop plans, continued.

Crop	Stage	Rate per acre	Method	Pack
Potato	Land Preparation	2 bags	Drill at sowing; broadcast if no drill	50 kg
Tomato	Land Preparation/Transplantation	1 bag	Broadcasting/Fertigation	50 kg
Tomato	Germination	1 bag	Broadcasting/Fertigation	50 kg
Chili	Land Preparation/Transplantation	2 bags	Broadcasting	50 kg
Onion	Land Preparation/Transplanting	1 bag	Broadcasting	50 kg
Citrus, 5 years and over	Land Preparation	1 bag	Broadcasting	50 kg

The rows continue from the previous slide. Each rate is per acre, and the crop page scales it to the size of the field.

WHAT VAN DOES NOT CLAIM FOR THIS PRODUCT

A dealer who makes any of these claims is speaking for himself, not for VAN.

1. That 70 to 80% of applied phosphate is fixed. That figure was a recovery statistic misread as soil chemistry, and VAN has withdrawn it.

2. A yield gain over DAP or TSP (triple superphosphate). The 1 wheat trial covers 1 season on 1 farm, and TSP was not in it.

3. A field-scale pH change, a soil-test change, or a percentage of phosphate saved.

4. That the humic seal keeps calcium away from the phosphate. It is the acidic zone that slows the reaction with calcium.

5. A humic content figure, or a mango flowering claim.

HOW A GROWER VERIFIES A BAG

Every batch is tested, and the batch number on the bag identifies its certificate.

1 · The batch number

Every bag carries the batch number. Read it to the grower and show him where it is printed.

2 · The certificate of analysis

On van.com.pk, a grower can read the certificate of analysis for any bag by entering the batch number printed on it. The grower does not need to post a sample.

3 · The laboratory

VAN Lab is accredited by the Pakistan National Accreditation Council (PNAC) to ISO/IEC 17025:2017, the international standard for testing laboratories, as LAB 336. Every batch is released through the laboratory before it ships. The certificate carries the laboratory's number.

What a certificate for Green Phosphate shows

The certificate reports the P_2O_5 and nitrogen measured in the batch, against the declared analysis 6-32-0. The bag carries the batch number, Pakistan Standard PS 933-2026 and licence CM/L-2719/2023. No humic content figure is on the bag, and none is given here. This training shows no batch result. A grower can read the certificate of his own batch on van.com.pk by its batch number.

THE URDU LINES · REVIEW STATUS UNDER EACH LINE

The lines a dealer says in Urdu.

The first line is the product's main message. Where its nutrient has one, the line gives VAN's promise word: the 1-word brand message VAN uses for that nutrient. The other 3 lines are key sentences of this deck. Each English line gives the meaning of the Urdu beside it. Send any correction to partner@van.com.pk.

Promise word REACH: "first the root, then everything".

پہلے جڑ، پھر سب کچھ

English meaning · Urdu reviewed 27 Sep 2026

Do not compare kilograms; look at what reaches the crop.

کلو کا موازنہ نہ کریں، وہ دیکھیں جو فصل تک پہنچے

English meaning · Urdu reviewed 27 Sep 2026

Drill it at sowing; broadcast it at land preparation if you cannot.

زمین کی تیاری پر ڈرل کریں، ڈرل نہ ہو تو چھٹہ

English meaning · Urdu line pending review

The acidic zone is around the granule, not across the whole field.

تیزابی حلقہ دانے کے گرد ہے، پورے کھیت میں نہیں

English meaning · Urdu reviewed 27 Sep 2026

TALK TO VAN

Use the address that matches your request.

For growers

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch or a dealer near you.

For the trade

partner@van.com.pk

Dealers, distributors and the trade: stock, training and corrections to this deck.

For everything else

info@van.com.pk

Supply, the laboratory or a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore, has 1 helpline and 3 addresses. VAN sends no group messages; every grower who writes gets a direct reply.

Green Phosphate goes into the soil before or at sowing. Do not use the kilograms to call one bag better.

